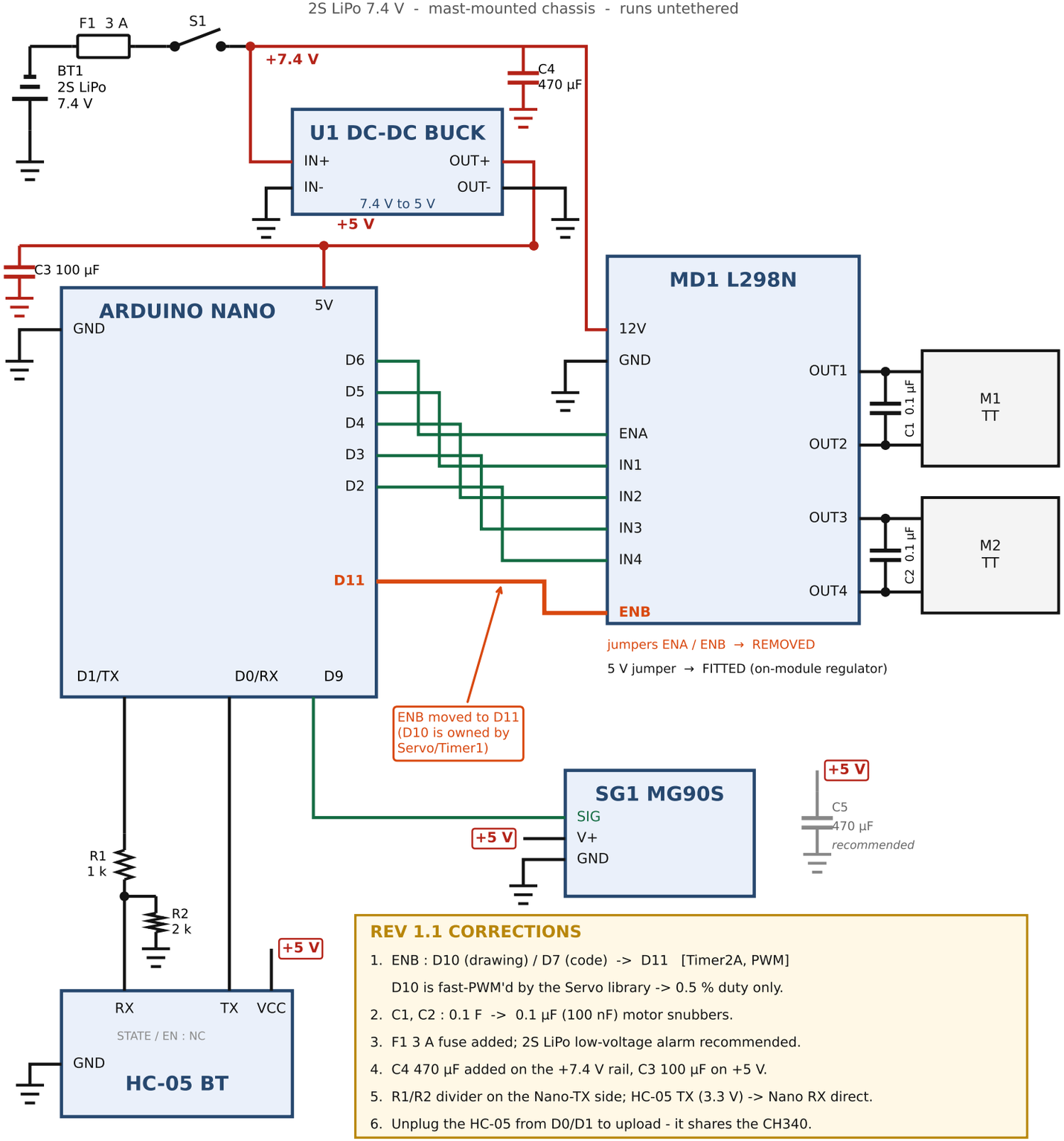
ROBOT

2S LiPo 7.4 V - mast-mounted chassis - runs untethered

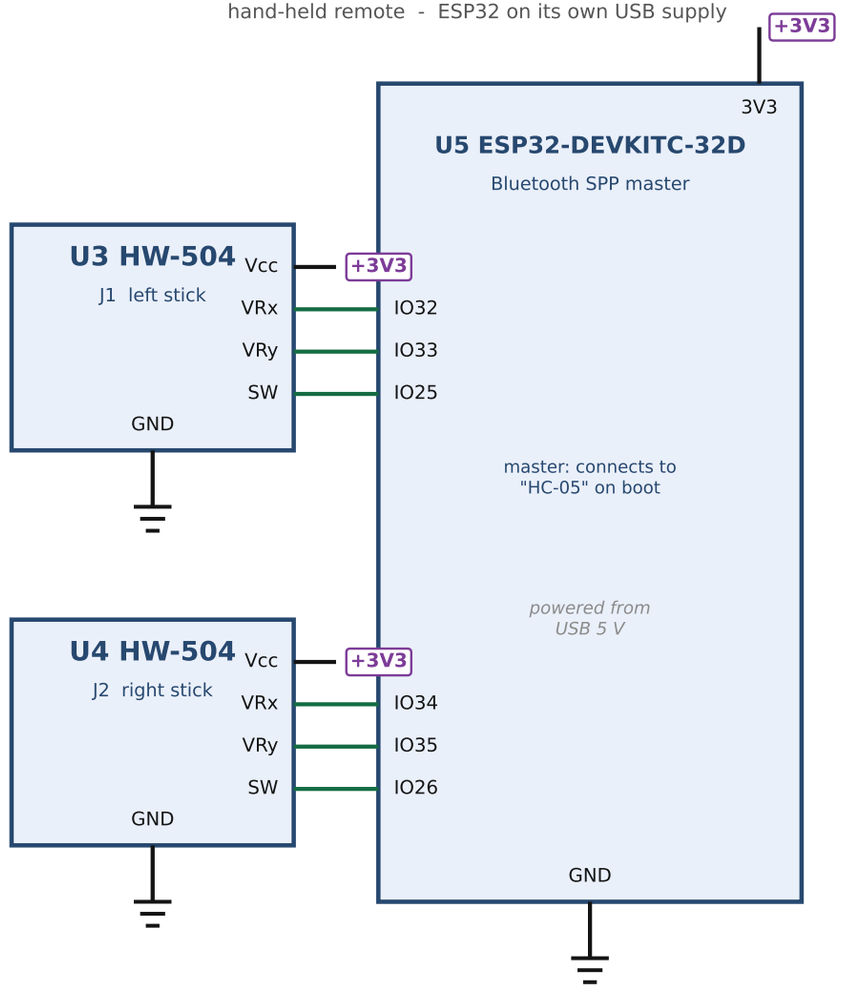


REV 1.1 CORRECTIONS

1. ENB : D10 (drawing) / D7 (code) -> D11 [Timer2A, PWM]
   D10 is fast-PWM'd by the Servo library -> 0.5 % duty only.
2. C1, C2 : 0.1 F -> 0.1 µF (100 nF) motor snubbers.
3. F1 3 A fuse added; 2S LiPo low-voltage alarm recommended.
4. C4 470 µF added on the +7.4 V rail, C3 100 µF on +5 V.
5. R1/R2 divider on the Nano-TX side; HC-05 TX (3.3 V) -> Nano RX direct.
6. Unplug the HC-05 from D0/D1 to upload - it shares the CH340.

CONTROLLER

hand-held remote - ESP32 on its own USB supply



CONTROLLER NOTES

HW-504 Vcc on +3V3 - GPIO34/35 are input-only and NOT 5 V tolerant.

SW pins need no resistors: firmware uses INPUT_PULLUP.

No 5 V rail on the remote - the ESP32 runs from its own USB supply.

Pick & Place Robot - Group 5	
REV 1.1 corrected	SHEET 1/1 REC Task 2